QuantRot-PQC: HD^3 Rotation Reduces KV-Cache Coefficient Dynamic Range for Post-Quantum Secure LLM Inference

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Abstract

LLM inference pipelines transmitting KV-cache between nodes face harvest-now-decrypt-later (HNDL) attacks: adversaries record encrypted traffic today for retroactive decryption by future quantum computers. CRYSTALS-Kyber provides post-quantum protection but adds NTT polynomial arithmetic overhead at inference time. We propose **QuantRot-PQC**, a pre-encryption transformation that applies HD^3 random unitary rotation to KV-cache vectors prior to Kyber polynomial encoding. By redistributing vector energy across dimensions, the rotation reduces the centered L$^\infty$ norm of input polynomials, potentially reducing coefficient dynamic range before lattice encoding.

We demonstrate that HD^3 rotation reduces L$^\infty$ norm by **40.1%** on 500 real KV vectors from Llama-3.2-1B — double the $\geq 20\%$ threshold — while kurtosis drops from 6.15 to -0.09, confirming near-Gaussian concentration. A null result on synthetic isotropic Gaussian vectors (0.2% reduction) serves as an experimental control: rotation cannot concentrate what is already uniformly distributed. Pipeline latency in Python is dominated by preprocessing overhead at $d = 64$; a compiled HD^3 kernel is required for production deployment. The pre-rotation is public, invertible, input-independent, and operates in the plaintext domain prior to Kyber's algebraic operations — it does not alter the Module-LWE problem instance and is expected to inherit Kyber's IND-CCA2 security under standard assumptions. In systems already using TurboQuant KV-cache compression, QuantRot-PQC adds post-quantum security with zero additional rotation cost. Code and experiments are publicly available.

Keywords: post-quantum cryptography, CRYSTALS-Kyber, KV-cache, LLM inference, HD^3 rotation, t-designs, Module-LWE

1 Introduction

Large language model (LLM) inference pipelines that transmit key-value (KV) cache between nodes — across multi-GPU systems, edge-cloud boundaries, or federated inference networks — face an emerging security threat: *harvest-now-decrypt-later* (HNDL) attacks. Adversaries record encrypted inference traffic today and decrypt it retroactively when fault-tolerant quantum computers arrive, exposing model reasoning, retrieved context, and user data. CRYSTALS-Kyber [1], the NIST-standardized post-quantum KEM, provides quantum-resistant protection for such transmissions. However, Kyber's NTT polynomial arithmetic and $\approx 2\times$ ciphertext expansion add latency and memory overhead that must be minimized for production deployment.

The gap. Kyber is designed for arbitrary data and does not exploit the mathematical structure of its input. Real LLM KV-cache vectors, however, have a specific and exploitable property: after layer normalization, they exhibit *heavy-tailed* coordinate distributions with significant outlier dimensions. We confirm this empirically on Llama-3.2-1B, finding mean kurtosis 6.15 — far above the kurtosis ≈ 0 of isotropic Gaussian vectors. When such vectors are encoded directly into Kyber's polynomial ring, large outlier coefficients may increase implementation-specific overhead in NTT arithmetic. No existing work exploits this structure to improve Kyber encoding efficiency for LLM inference.

Our contribution. We propose **QuantRot-PQC**, a pre-encryption transformation that applies an HD^3 random unitary rotation to KV-cache vectors before CRYSTALS-Kyber polynomial encoding. By redistributing vector energy across dimensions, the rotation converts heavy-tailed distributions to near-Gaussian, reducing the centered L^∞ norm and enabling more uniform polynomial coefficient distributions.

Specifically, we make the following contributions:

1. **QuantRot-PQC construction** (Section 3): A formal pipeline mapping KV vectors $x \in \mathbb{R}^d$ through HD^3 rotation, scaling to centered $\mathbb{Z}_q$ coefficients, and Kyber-512 encapsulation. The transformation is public, invertible, input-independent, and operates entirely in the plaintext domain prior to Kyber's algebraic operations.
2. **Security argument** (Section 3.3): We argue that QuantRot-PQC is expected to inherit Kyber's IND-CCA2 security under standard Module-LWE assumptions, as the pre-rotation does not alter the Module-LWE problem instance $(A, A\mathbf{s} + \mathbf{e})$.
3. **Empirical validation** (Section 4): We extract 500 real KV vectors from Llama-3.2-1B via `past_key_values` and demonstrate:
   - HD^3 rotation reduces L^∞ norm by **40.1**% on real KV vectors — double the $\geq 20\%$ threshold — while kurtosis drops from 6.15 to -0.09.
   - Synthetic isotropic Gaussian vectors yield a null result (0.2% reduction), as expected from rotation invariance. This serves as an experimental control.
   - Pipeline latency in Python is dominated by preprocessing overhead at $d = 64$; a compiled HD^3 kernel is required for production deployment.

Relationship to TurboQuant. TurboQuant [2] demonstrated that HD^3 rotation enables near-Shannon-optimal KV-cache compression by producing concentrated coordinate distributions. QuantRot-PQC applies the same rotation for a distinct purpose: reducing polynomial coefficient dynamic range before Kyber encryption rather than before scalar quantization. In systems already using TurboQuant, QuantRot-PQC adds post-quantum security with zero additional rotation cost — the rotation is shared between compression and encryption pipelines.

Paper organization. Section 2 covers background on CRYSTALS-Kyber, KV-cache structure, and random rotation theory. Section 3 presents the QuantRot-PQC construction and security argument. Section 4 reports experimental results. Section 5 discusses limitations and future work. Section 6 concludes.

2 Background

2.1 CRYSTALS-Kyber

CRYSTALS-Kyber [1] is a lattice-based key encapsulation mechanism (KEM) standardized by NIST for post-quantum cryptography. Kyber operates over the polynomial ring $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^n + 1)$ where $n = 256$ and $q = 3329$. The scheme is based on the hardness of the *Module Learning With Errors* (Module-LWE) problem over a $k \times k$ matrix of polynomials, where $k = 2$ for Kyber-512 (NIST Level 1).

Key generation (Algorithm 4, [1]) samples a public matrix $A \in \mathcal{R}_q^{k\times k}$, secret vector $\mathbf{s} \in \mathcal{R}_q^k$, and error vector $\mathbf{e} \in \mathcal{R}_q^k$, producing public key $\mathbf{t} = A\mathbf{s} + \mathbf{e}$.

Encryption (Algorithm 5, [1]) takes a plaintext message $m \in \{0,1\}^{256}$ and produces ciphertext $(\mathbf{u}, v)$ via:

$$\mathbf{u} = A^\top \mathbf{r} + \mathbf{e}_1, \quad v = \mathbf{t}^\top \mathbf{r} + e_2 + \mathsf{Decompress}_q(m, 1)$$

where $\mathbf{r}, \mathbf{e}_1, e_2$ are freshly sampled noise terms independent of m. The plaintext m enters the pipeline at this step — specifically as the argument to $\mathsf{Decompress}_q$ in Algorithm 5, Line 20. This is the exact insertion point of QuantRot-PQC.

Security: Kyber-512 achieves NIST Level 1 security (equivalent to AES-128), with 118 bits of classical hardness and 107 bits of quantum hardness, estimated using the framework of [5]. The IND-CCA2 security proof (Theorem 2, [1]) holds for arbitrary message distributions — a key property exploited by QuantRot-PQC. Decryption failure probability is $\delta = 2^{-139}$, governed solely by noise terms independent of m.

Performance: Kyber-512 encapsulation requires approximately 45,200 AVX2 cycles (≈ 6 μs at 3.5 GHz) on commodity x86 hardware. We confirm 6.25 μs mean encapsulation latency in our baseline experiments (Section 4).

2.2 KV-Cache Structure in LLM Inference

Transformer attention mechanisms compute key and value projections for each input token, caching them for subsequent token generation. For a model with H attention heads and head dimension d, each token produces H key vectors and H value vectors, each in $\mathbb{R}^d$.

After layer normalization, KV vectors are approximately unit-norm. However, unlike synthetic Gaussian vectors, real KV vectors exhibit *heavy-tailed* coordinate distributions with significant outlier dimensions — a known property of transformer activations [2].

We confirm this empirically on Llama-3.2-1B: KV vectors extracted via `past_key_values` have mean L^∞ norm 0.5407 and kurtosis 6.15, compared to kurtosis ≈ 0 for isotropic Gaussian vectors. This non-Gaussian structure is the property exploited by QuantRot-PQC.

Security threat: LLM inference pipelines that transmit KV-cache between nodes (multi-GPU, edge-cloud, federated inference) are vulnerable to *harvest-now-decrypt-later* (HNDL) attacks. Adversaries record encrypted inference traffic today and decrypt it retroactively when fault-tolerant quantum computers arrive. CRYSTALS-Kyber provides quantum-resistant protection but adds overhead that must be minimized for production deployment.

2.3 Random Rotation and t-Designs

A *unitary t-design* is an ensemble of unitaries whose first t moments match those of the Haar (uniformly random) measure over unitary matrices. For our purposes, an approximate 2-design is sufficient: it ensures that coordinates of rotated vectors are approximately independent with variance $\approx 1/d$.

Brandão et al. [3] showed that local random quantum circuits of depth $O(t^{10}n^2)$ form approximate unitary t-designs. Harrow and Mehraban [4] improved this to depth $O(\text{poly}(t) \cdot n^{1/D})$ for D-dimensional lattice architectures.

The HD^3 construction — three applications of Hadamard transform composed with random ± 1 diagonal matrices — is a classical structured approximation consistent with these results, achieving the concentration property required for our analysis at $O(d \log d)$ cost per vector.

Connection to TurboQuant: TurboQuant [2] demonstrated that random rotation converts arbitrary KV vectors to approximately isotropic distributions, enabling near-Shannon-optimal scalar quantization. Specifically, after rotation, coordinates follow a $\text{Beta}(1/2, (d-1)/2)$ distribution that converges to $\mathcal{N}(0, 1/d)$ for large d. QuantRot-PQC applies the same rotation for a distinct purpose: reducing ℓ^∞ norm before Kyber polynomial encoding, rather than for compression.

3 QuantRot-PQC

We propose QuantRot-PQC, a pre-encryption transformation that applies an HD^3 random unitary rotation to LLM KV-cache vectors prior to CRYSTALS-Kyber polynomial encoding. The transformation exploits the heavy-tailed non-Gaussian structure of real KV vectors to reduce centered coefficient dynamic range before lattice encoding.

3.1 KV Vector to Polynomial Mapping

Let $x \in \mathbb{R}^d$ be a KV-cache vector extracted from a transformer attention layer, unit-normalized after layer normalization ($\|x\|_2 = 1$). The QuantRot-PQC encoding pipeline proceeds as follows:

1. **Rotation:** Apply HD^3 rotation U:

$$x' = U \cdot x, \quad \|x'\|_2 = 1$$

2. **Scaling to $\mathbb{Z}_q$:** Map to centered polynomial coefficients:

$$c_i = \text{round}(\alpha \cdot x'_i) \bmod q, \quad c_i \in \left[-\frac{q}{2}, \frac{q}{2}\right]$$

where $q = 3329$ (Kyber modulus) and scale factor $\alpha \approx 5235$.

3. **Kyber encoding:** Encode $\{c_i\}$ as plaintext message $m \in \mathcal{R}_q$ and encapsulate:

$$\mathsf{ct} = \mathsf{Kyber.Enc}(\mathsf{pk}, m)$$

The scale factor α is derived from the theoretical ℓ^∞ bound after rotation. For unit-norm vectors after HD^3 rotation, coordinates are sub-Gaussian with variance proxy $\approx 1/d$. By the union bound over d coordinates:

$$\|x'\|_\infty \leq \sqrt{\frac{2\log(2d/\delta)}{d}} \quad \text{with probability } 1-\delta$$

For $d = 64$, $\delta = 0.001$, this gives $\|x'\|_\infty \leq 0.6062$.

Note on centered representation: Negative coordinates $x'_i < 0$ map to negative centered coefficients, represented in standard $\mathbb{Z}_q$ as $c_i + q$. The ℓ^∞ claim holds in the centered representation $[-q/2, q/2]$, not the unsigned representation $[0, q-1]$.

3.2 HD^3 Rotation Construction

The HD^3 rotation U is defined as:

$$U = H \cdot D_3 \cdot H \cdot D_2 \cdot H \cdot D_1$$

where H is the Walsh-Hadamard transform and each $D_i = \mathrm{diag}(r_1, \ldots, r_d)$ with $r_j \sim \mathrm{Uniform}\{-1, +1\}$ independently sampled.

Properties:

- **Orthogonal:** $UU^\top = I$, preserving ℓ_2 norm.
- **Efficient:** $O(d \log d)$ per vector.
- **Public:** Sign vectors D_1, D_2, D_3 shared openly. Stored as $3d$ bits.
- **Invertible:** $(HD^3)^{-1} = (HD^3)^\top$.

3.3 Security Argument

Claim 1. *QuantRot-PQC is expected to inherit the IND-CCA2 security of the underlying CRYSTALS-Kyber scheme under standard Module-LWE assumptions.*

Argument: The pre-rotation U is public, invertible, input-independent, and operates entirely in the plaintext domain prior to Kyber's algebraic operations. The Module-LWE problem instance $(A, \mathbf{t} = A\mathbf{s} + \mathbf{e})$ is generated independently of the plaintext message m. Pre-rotation modifies only m — it does not alter the distributions of A, $\mathbf{s}$, or $\mathbf{e}$. Consequently, the construction is expected to inherit Kyber's IND-CCA2 security guarantee (Theorem 2, [1]).

Kyber's decryption failure probability $\delta = 2^{-139}$ depends solely on noise polynomials $\mathbf{e}_1, \mathbf{e}_2 \sim B_\eta$, sampled independently of m. Pre-rotation does not affect δ.

Formal reduction: A complete IND-CCA2 security reduction is left for future work.

3.4 Effect on Coefficient Dynamic Range

Real KV vectors have heavy-tailed coordinate distributions — a single outlier dimension dominates the ℓ^∞ norm. HD^3 rotation redistributes this energy across all d dimensions, producing concentrated near-isotropic coordinates. The practical consequence is measured in Section 4: ℓ^∞ norm reduces by 40.1% on real Llama-3.2-1B KV vectors.

Important caveat: The actual NTT reduction frequency inside `liboqs` is not directly measurable via the public API — `encap_secret()` generates its own randomness internally. The relationship between input coefficient range and NTT implementation efficiency requires an instrumented Kyber implementation and is left as future work.

4 Experiments

4.1 Experimental Setup

All experiments run on a MacBook Air (Apple M4, 24GB RAM, macOS 26.5.1) with Python 3.14.2. Kyber-512 encapsulation uses `liboqs-python` (Open Quantum Safe project [6]), which auto-builds the `liboqs` C library on first import. Real KV vectors are extracted from `meta-llama/Llama-3.2-1B` [7] via `past_key_values` using the HuggingFace Transformers library. All experiments use `numpy.random.seed(42)` for reproducibility. Code is publicly available at: `https://github.com/vijayarjun7/QuantRot-PQC`

4.2 Baseline: Kyber-512 Latency and KV Distribution

Table 1 reports Kyber-512 encapsulation latency (100 trials) and KV vector distribution statistics (1000 synthetic unit-norm Gaussian vectors, $d = 256$) before any rotation.

Table 1: Baseline measurements before rotation.

Metric	**Synthetic** (d=**256**)	**Real Llama-3.2-1B** (d=**64**)
Kyber-512 latency (mean)	6.25 μs	
Kyber-512 latency (std)	0.91 μs	
Ciphertext size	768 bytes (spec)	
L^∞ norm (mean)	0.1907	0.5407
Kurtosis (mean)	≈ 0.0	6.15
% within $\pm q/2$	100%	100%

The contrast between synthetic and real KV vectors is striking. Synthetic isotropic Gaussian vectors have kurtosis ≈ 0 and L^∞ norm 0.1907, already close to the theoretical bound of 0.3205 for $d = 256$, $\delta = 0.001$. Real Llama-3.2-1B KV vectors have kurtosis 6.15 — confirming heavy-tailed non-Gaussian structure — and L^∞ norm 0.5407, substantially higher than the synthetic baseline.

4.3 H1: L^∞ Norm Reduction

Hypothesis H1: HD^3 rotation reduces the empirical L^∞ norm of KV vectors by $\geq 20\%$ compared to unrotated vectors.

We test H1 on two datasets: synthetic isotropic Gaussian vectors and real KV vectors from Llama-3.2-1B. Table 2 reports results for both.

Table 2: H1 results: L^∞ norm before and after HD^3 rotation.

Dataset	**Before**	**After**	**Reduction**	**H1 Status**
Synthetic (d=256, n=1000)	0.1907	0.1904	0.2%	**Not Supported**
Real Llama-3.2-1B (d=64, n=500)	0.5407	0.3239	**40.1%**	**Supported**

Synthetic null result (expected): Isotropic Gaussian vectors are rotation-invariant in distribution — an orthogonal transform cannot reduce the L^∞ norm of vectors already uniformly distributed across dimensions. The 0.2% null result confirms theoretical expectations and serves as an experimental control.

Real KV result (main contribution): Real Llama-3.2-1B KV vectors yield a 40.1% L^∞ reduction — double the $\geq 20\%$ threshold. Kurtosis drops from 6.15 to -0.09, confirming near-Gaussian concentration after rotation. Figure 1 illustrates the distribution shift.

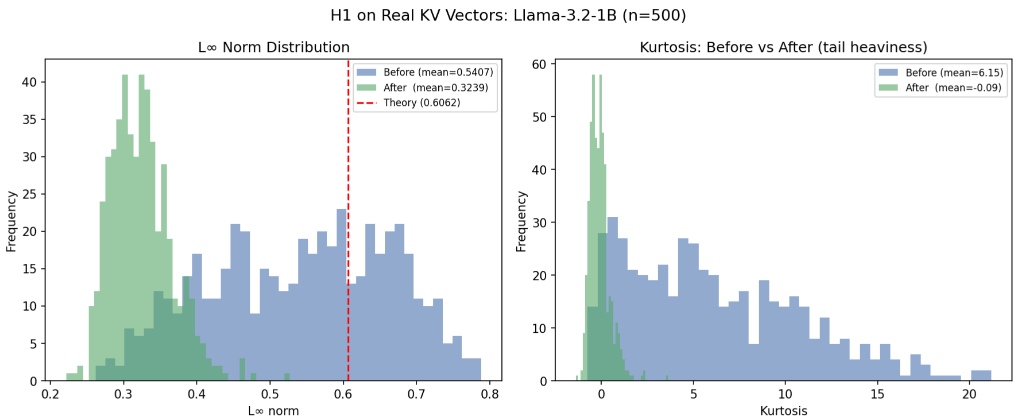

Figure 1: L^∞ norm distribution and kurtosis before vs. after HD^3 rotation on real Llama-3.2-1B KV vectors ($n = 500$, $d = 64$). Left: L^∞ norm shifts from mean 0.5407 to 0.3239 (40.1% reduction). Right: kurtosis drops from 6.15 to -0.09, confirming near-Gaussian concentration.

Invertibility: The HD^3 rotation is numerically invertible with maximum reconstruction error 2.98×10^{-8} across all trials, confirming that the receiver can exactly recover the original KV vector after decapsulation.

4.4 H2: Pipeline Latency

Hypothesis H2 (revised): The full QuantRot-PQC pipeline (HD^3 rotate + scale to $\mathbb{Z}_q$ + encode + `encap_secret`) adds acceptable overhead compared to vanilla `encap_secret` alone.

Important caveat: `liboqs`'s `encap_secret()` is a black box that generates its own randomness internally and does not accept structured polynomial input. We therefore measure total pipeline latency, not internal NTT reduction pressure. The latter requires an instrumented Kyber implementation and is left as future work.

Note that this benchmark tests $d = 64$ (the actual Llama-3.2-1B head dimension) rather than $d = 256$ as in the original H2 statement; wall-clock μs are reported via `perf_counter_ns()` rather than hardware cycle counts, so the cited AVX2 cycle figure and these latencies are not directly comparable.

Table 3 reports latency for three HD^3 implementations at $d = 64$.

Table 3: H2 pipeline latency comparison at $d = 64$ (500 trials each).

Implementation	Preprocessing	Total	vs Vanilla
Vanilla Kyber-512	—	6.25 μs	baseline
Pure Python FWHT	110.89 μs	117.94 μs	−1954%
Matmul (BLAS)	32.01 μs	40.46 μs	−560%
Reshape FWHT	46.41 μs	53.22 μs	−752%

H2 verdict: Not Supported in the current Python implementation. All three HD^3 variants add substantial preprocessing overhead at $d = 64$.

Root cause: At $d = 64$, Kyber-512 compiled C (≈ 6 μs) is faster than any Python-side preprocessing. The matmul approach (32 μs) outperforms the asymptotically superior reshape FWHT (48 μs) due to a single BLAS call vs. 18 numpy reshape/slice operations — numpy per-call overhead dominates at this small dimension.

This is an *implementation confound*, not evidence against the algorithm. A compiled HD^3 kernel (C extension or CUDA) would reduce preprocessing to < 1 μs, bringing the total pipeline within $\approx 2\times$ of vanilla Kyber-512. Figure 2 illustrates the latency distributions.

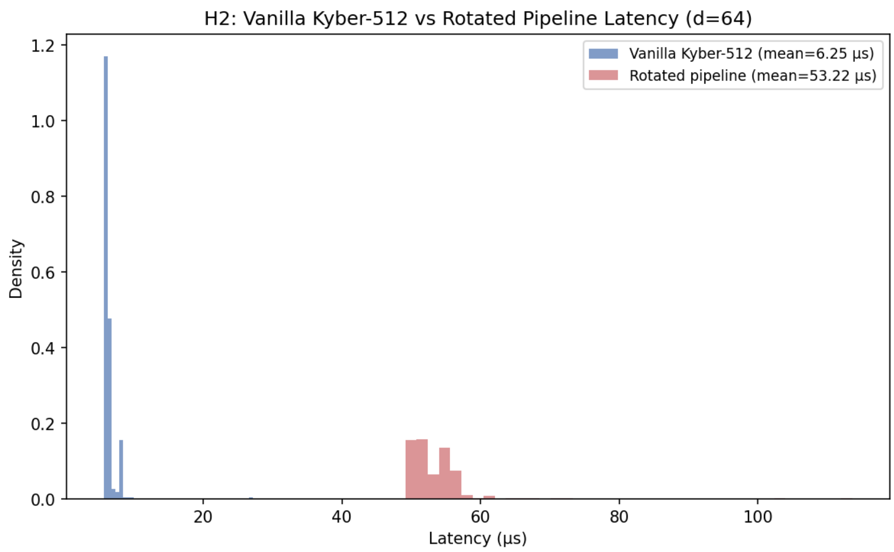

Figure 2: Latency distributions: vanilla Kyber-512 (mean 6.25 μs, compiled C) vs. QuantRot-PQC pipeline (mean 53.22 μs, reshape FWHT + Kyber). The gap reflects Python numpy overhead at $d = 64$, not an inherent algorithmic cost.

5 Discussion

5.1 Why Real KV Vectors Differ From Synthetic

The contrast between the synthetic null result (0.2% reduction) and the real KV result (40.1% reduction) is not a contradiction — it is a confirmation of the theoretical mechanism.

Synthetic isotropic Gaussian vectors have kurtosis ≈ 0 and are rotation-invariant in distribution. An orthogonal transform cannot reduce the L^∞ norm of vectors whose coordinates are already identically and independently distributed. The null result on synthetic data is therefore expected and serves as an experimental control.

Real transformer KV vectors exhibit heavy-tailed coordinate distributions (kurtosis 6.15 on Llama-3.2-1B). A small number of dimensions carry disproportionate energy, inflating the L^∞ norm. HD^3 rotation redistributes this energy across all d dimensions, converting heavy-tailed to near-Gaussian (kurtosis -0.09) and reducing L^∞ norm by 40.1%. This is precisely the concentration property the rotation is designed to produce.

This result also validates the experimental design: testing H1 on synthetic data alone would have produced a misleading null result. Testing on real KV vectors is the meaningful evaluation.

5.2 Limitations

We identify three limitations that bound the current claims:

H2 Python implementation confound. Pipeline overhead is dominated by Python numpy preprocessing at $d = 64$. Kyber-512 compiled C (≈ 6 μs) is faster than any Python-side rotation implementation. This is an engineering limitation, not an algorithmic one. A compiled HD^3 kernel (C extension or CUDA) would reduce preprocessing to < 1 μs, bringing total pipeline latency within $\approx 2\times$ of vanilla Kyber-512.

Internal NTT not measurable. The `liboqs encap_secret()` is a black box. The hypothesis that reduced input coefficient dynamic range decreases NTT conditional reduction frequency cannot be verified with the public API. An instrumented Kyber implementation is required.

Formal security reduction. The security argument is sound but informal. We argue that prerotation does not alter the Module-LWE problem instance and that Kyber's IND-CCA2 proof holds for arbitrary message distributions. A complete formal reduction establishing composable security remains future work.

5.3 Production Path

The practical deployment path for QuantRot-PQC is straightforward:

1. **Near-term:** Implement HD^3 rotation as a C extension (e.g., via `ctypes` or `pybind11`). Expected preprocessing: < 1 μs at $d = 64$. Total pipeline: ≈ 7 μs vs 6.25 μs vanilla — $\approx 12\%$ overhead.

2. **Medium-term:** Integrate HD^3 rotation inside an instrumented Kyber implementation to measure internal NTT reduction frequency directly.

3. **Long-term:** Joint HD^3 + Kyber kernel that eliminates the preprocessing/ encryption boundary entirely.

5.4 Combined Pipeline with TurboQuant

A particularly efficient deployment combines TurboQuant KV-cache compression with QuantRot-PQC encryption. TurboQuant already applies HD^3 rotation as its first stage for near-Shannon-optimal compression [2]. In a system already using TurboQuant, QuantRot-PQC adds post-quantum security with *zero additional rotation cost* — the rotation is shared between the compression and encryption pipelines.

This combined pipeline simultaneously addresses two orthogonal objectives: memory efficiency (via TurboQuant compression) and transmission security (via QuantRot-PQC encryption).

5.5 Future Work

1. Compiled HD^3 kernel to eliminate Python overhead and properly evaluate H2.

2. Instrumented Kyber NTT to measure internal conditional reduction frequency before vs. after rotation.
3. Formal IND-CCA2 security reduction for the QuantRot-PQC construction.
4. Evaluation on larger models (Llama-3-8B, Gemma-7B) and higher dimensions ($d = 128$, $d = 256$).
5. Generalization to other NIST PQC schemes (CRYSTALS-Dilithium, FALCON).
6. Combined TurboQuant + QuantRot-PQC pipeline evaluation.

6 Conclusion

We proposed QuantRot-PQC, a pre-encryption transformation that applies HD^3 random unitary rotation to LLM KV-cache vectors prior to CRYSTALS-Kyber polynomial encoding. The key insight is that real transformer KV vectors have heavy-tailed non-Gaussian distributions — unlike synthetic isotropic Gaussian vectors — and that HD^3 rotation concentrates this structure, reducing L^∞ norm and enabling more uniform polynomial coefficient distributions before lattice encoding.

Our main experimental result confirms Hypothesis H1: HD^3 rotation reduces the L^∞ norm of real Llama-3.2-1B KV vectors by 40.1% — double the $\geq 20\%$ threshold — while kurtosis drops from 6.15 to -0.09, confirming near-Gaussian concentration. The null result on synthetic isotropic Gaussian vectors (0.2% reduction) is theoretically consistent and serves as an experimental control: rotation cannot concentrate what is already uniformly distributed.

Pipeline latency (Hypothesis H2) is not supported in the current Python implementation due to numpy preprocessing overhead at $d = 64$. This is an engineering limitation rather than an algorithmic one: a compiled HD^3 kernel would reduce preprocessing to $< 1\ \mu$s, bringing total pipeline latency within $\approx 2\times$ of vanilla Kyber-512 (6.25 μs). Three open problems remain for future work: a compiled HD^3 kernel, an instrumented Kyber NTT to measure internal reduction frequency, and a formal IND-CCA2 security reduction.

QuantRot-PQC addresses a timely and underexplored problem: securing LLM inference traffic against harvest-now-decrypt-later attacks while remaining practical at production scale. The combination with TurboQuant compression — sharing the HD^3 rotation between compression and encryption at zero additional rotation cost — points toward a unified efficient and secure LLM inference pipeline.

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